Case Studies in Operations Management

Mathematical Model Milestone 1

Business and Logistics Plan for "Fresh Veggie" (FV)

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1 Model Assumptions

The model corresponds to a first-stage strategic network design. It evaluates a set of predefined facility configurations and selects the one that maximizes estimated annual profit.

The following simplifying assumptions are adopted:

- Demand is constant over the year and derived from per-capita consumption.
- Each facility configuration $z \in \mathcal{Z}$ has a precomputed coverage level $p(z)$ measured in number of people.
- The company can serve at most a fraction α^{max} of the covered population.
- Revenue is estimated using per-capita annual demand and average selling prices.
- Only fixed opening costs of facilities are considered at this stage.

2 Sets

- F: set of all candidate facilities.
- $\mathcal{Z}$: set of all feasible facility configurations, where each $z \in \mathcal{Z}$ is a binary vector indicating which facilities are active.
- K: set of products:

$$K = \{\text{Roma Tomatoes}, \text{Cauliflower}, \text{Broccoli}, \text{Asparagus}, \text{Green Bell Pepper}\}$$

3 Parameters

- $p(z)$: number of people covered by configuration $z \in \mathcal{Z}$.
- α^{max}: maximum fraction of covered demand that can be served.
- $D_k^{pc,ann}$: annual per-capita demand of product $k \in K$, expressed in kg/person/year.
- P_k: selling price per kilogram of product $k \in K$, expressed in \$/kg.
- $b_{f,z}$: binary parameter equal to 1 if facility $f \in F$ is included in configuration z, and 0 otherwise.
- C_f^{open}: fixed cost of opening facility $f \in F$, expressed in \$.

4 Decision Variables

- $y_z \in \{0, 1\}$: binary variable equal to 1 if configuration $z \in \mathcal{Z}$ is selected, and 0 otherwise.

5 Objective Function

The objective is to maximize the estimated annual profit associated with the selected facility configuration.

$$\max \sum_{z \in \mathcal{Z}} \left[p(z)\, \alpha^{max} \sum_{k \in K} P_k D_k^{pc,ann} - \sum_{f \in F} C_f^{open}\, b_{f,z} \right] y_z$$

6 Constraints

6.1 Configuration Selection

Exactly one facility configuration must be selected:

$$\sum_{z \in \mathcal{Z}} y_z = 1$$

6.2 Binary Variables

$$y_z \in \{0, 1\}, \quad \forall z \in \mathcal{Z}$$